AIRBNB NEW YORK CITY 2019

Exploratory Data Analysis and Linear Regression

PROJECT REPORT

Field	Details
Student name	Lipsa Priyadarshini
Department	Computer Science and Engineering (CSE)
Project dataset	AB_NYC_2019.csv
Date	26 October 2026
Dataset source	Airbnb NYC 2019 (public dataset; original download link to be added by student)

2. Objective

The project aims to inspect and understand New York City Airbnb listings from 2019, identify data quality issues, summarize numerical and categorical features, explore potential associations with listing prices, and prepare a baseline Linear Regression model for price prediction.

Analysis questions: How are listings distributed across room types and neighborhoods? Which variables may be related to price? How much missing information is present? What precautions are needed when interpreting a linear model?

3. Dataset Overview

Dataset property	Result
File name	AB_NYC_2019.csv
Number of rows	48,895
Number of columns	16
Unit of observation	One Airbnb listing
Train/test split	80% / 20%
Prediction algorithm	Linear Regression

The dataset commonly includes listing identifiers, host information, geographic attributes, room type, price, minimum nights, review counts, availability and review dates. The exact 16-column schema should be checked against the submitted CSV because versions may differ.

Representative features: price (numeric target), latitude/longitude (numeric), room_type and neighbourhood_group (categorical), number_of_reviews and availability_365 (numeric), last_review (date-like), reviews_per_month (numeric).

Sample records: Not reproduced here because the original CSV was not attached; paste a screenshot of `df.head()` from the completed notebook before final submission.

4. Data Quality and Cleaning

Check	Result / proposed action
Missing last_review	10,052 rows (20.56%); may indicate listings with no reviews.
Missing reviews_per_month	10,052 rows (20.56%); consider 0 where logically appropriate, with justification.
Duplicates	Check full-row and listing-ID duplicates; actual count not provided.
Outliers	Inspect price, minimum_nights and review variables using boxplots/IQR; avoid automatic deletion.
Data types	Convert date fields with error handling; encode categories for regression.
Leakage prevention	Fit imputation and encoding steps on training data only.

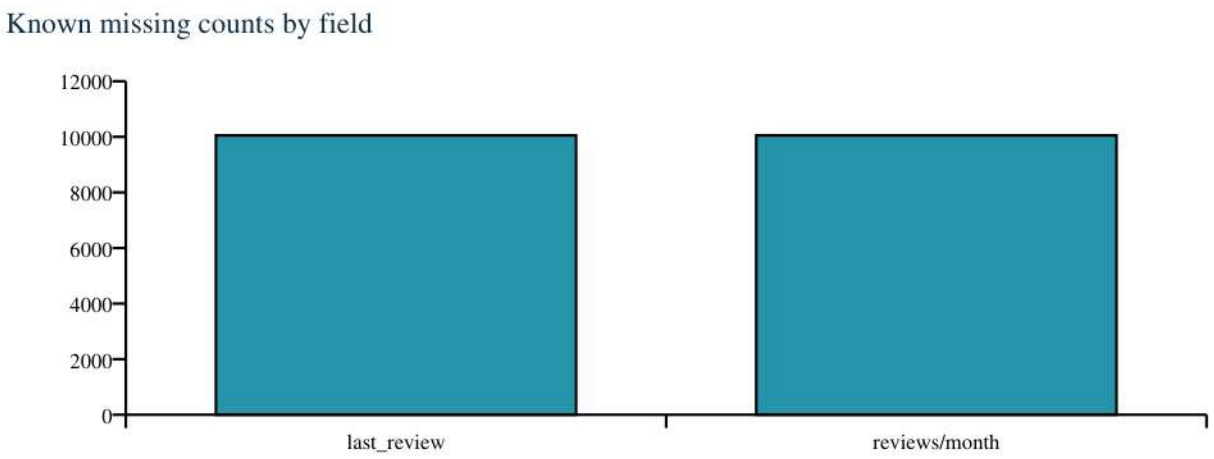
Missing values in the two review fields affect roughly one in five records. A missing last-review date should not be interpreted as a known date. Extreme prices may represent luxury listings or errors and must be investigated before deciding whether to filter them.

5. Descriptive Analysis

Descriptive analysis should include **count, mean, median, standard deviation, minimum, quartiles and maximum** for price, minimum_nights, number_of_reviews and availability_365. For categorical fields, include value counts and percentages for room_type and neighbourhood_group.

The source CSV and notebook outputs were not supplied, so exact price summaries, room-type frequencies and geographic totals are intentionally not fabricated. Insert your notebook tables here for a complete evidence-based submission.

Data completeness overview

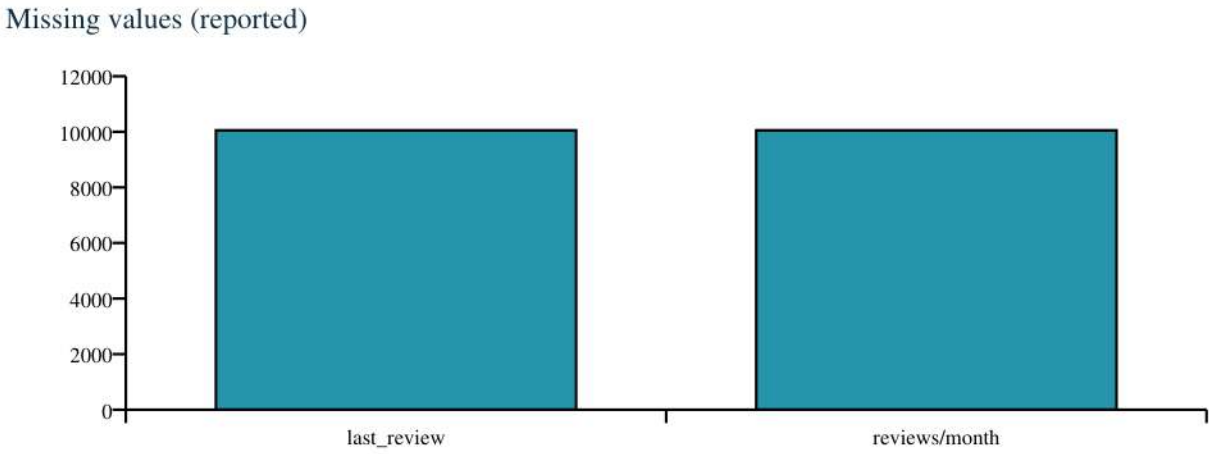


Observation: Both fields have the same reported missing count, but this alone does not prove that the missing rows are identical.

6. Visual EDA

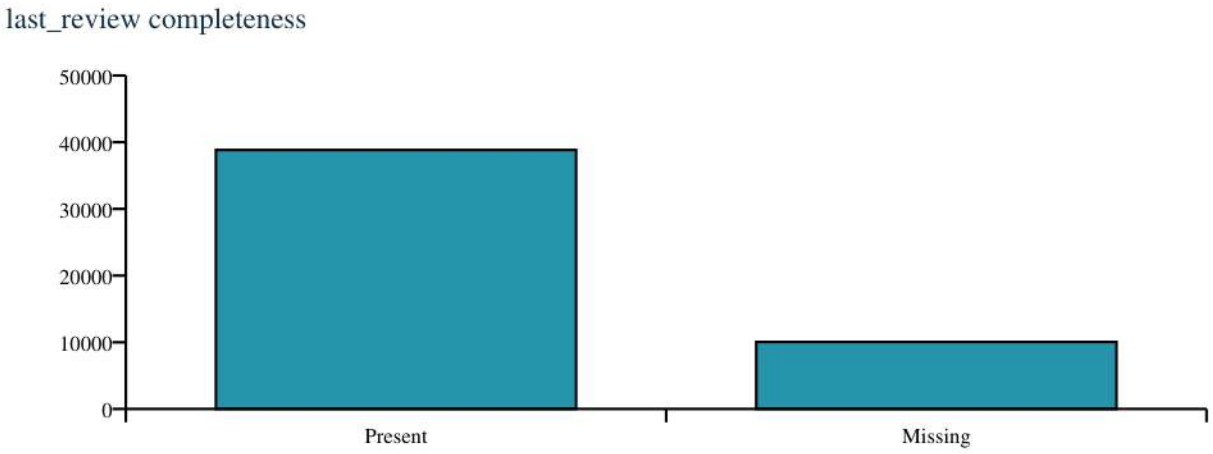
The project instructions require at least six titled and labeled charts with observations. Below are six chart specifications and interpretation prompts. Only charts directly supported by the supplied counts are plotted; the remaining charts should be inserted as actual notebook screenshots, not estimated graphics.

Chart 1. Missing values by column



Observation: Each field is missing in 10,052 of 48,895 listings (20.56%). Other columns require verification from the CSV.

Chart 2. Missing vs non-missing last_review



Observation: 38,843 entries (79.44%) are non-missing and 10,052 (20.56%) are missing.

Chart 3. Price distribution (histogram)

Insert notebook chart: histogram of price, ideally with a sensible x-axis limit or log scale for visibility. **Observation to verify:** describe skewness, concentration and unusually high values using the plotted output.

Chart 4. Room type distribution (count plot)

Insert notebook chart: count of listings by room_type. **Observation to verify:** identify the most common category and compare counts.

Chart 5. Price by neighborhood group (box plot)

Insert notebook chart: neighborhood group on x-axis and price on y-axis. **Observation to verify:** compare medians, spread and outliers.

Chart 6. Numeric-variable correlations (heatmap)

Insert notebook chart: correlation matrix for numeric columns after checking data types. **Observation to verify:** describe the strongest positive/negative correlations with price without claiming causation.

Additional recommended plots

A scatter plot of price versus minimum_nights or number_of_reviews and a geographical scatter plot of latitude versus longitude can strengthen the bivariate and multivariate discussion.

7. Linear Regression: Model Workflow

The target variable is **price**. Select numeric and properly encoded categorical predictors, split the data into **80% training** and **20% testing**, then fit a Linear Regression baseline. With 48,895 records, the usual scikit-learn split with test_size=0.2 gives approximately 39,116 training and 9,779 testing rows (before any row removal).

Stage	What to document
Preprocessing	Missing-value strategy, categorical encoding, outlier handling and feature selection.
Training	Fit LinearRegression on the training set only.
Evaluation	Report MAE, MSE, RMSE and R-squared on the held-out test set.
Diagnostics	Compare actual vs predicted price and inspect residuals.

Model results: Evaluation values were not provided. Add your actual notebook outputs here; no accuracy, R-squared or error values should be claimed without running the model.

8. Key Findings

Finding 1: The dataset contains 48,895 records and 16 columns, providing a substantial cross-sectional view of Airbnb listings.

Finding 2: last_review has 10,052 missing entries (20.56%).

Finding 3: reviews_per_month also has 10,052 missing entries (20.56%).

Finding 4: The chosen 80/20 split reserves a separate test set to assess generalization.

Finding 5: Linear Regression is an interpretable baseline, but model suitability must be judged using measured test errors and residual plots.

Limitations: Missing review information, possible extreme prices, categorical encoding choices and the snapshot nature of 2019 data can influence interpretation. A linear model may not capture complex nonlinear pricing patterns.

9. Conclusion

This project establishes a structured workflow for Airbnb NYC 2019 data auditing, descriptive analysis, visualization and price prediction with Linear Regression. The known data-quality issue is the missing information in the two review-related fields. The 80/20 split provides a standard basis for evaluating a baseline regression model.

The next stage is to attach the actual six visualizations, summary statistics, sample records and regression evaluation metrics from the completed Colab notebook. These outputs are necessary to substantiate conclusions about prices, neighborhoods, room types and model performance.

Future Scope

Possible extensions include comparing Ridge, Lasso and tree-based regressors; using cross-validation; engineering geographic features; and evaluating errors separately across room types or neighborhood groups.

Submission checklist

Required item	Status
Cover page, objective and dataset overview	Included
Missing-data audit and cleaning discussion	Included; some checks require notebook results
Descriptive statistics with actual outputs	To insert
Six real EDA charts and observations	Two count-based charts included; four to insert
Linear Regression test metrics	To insert
Key findings, limitations and conclusion	Included

Note: This report is a draft suitable for completing with evidence from the notebook. It should not be submitted as a fully verified analysis until the missing outputs have been added.